

Question	Answer	Marks
1(a)	angle (subtended at the centre of a circle) when arc (length) = radius	B1
1(b)(i)	$v = r\omega$	C1
	radius = $2.4 / 0.75$ = 3.2 m	A1
1(b)(ii)	$\omega = 2\pi f$	C1
	frequency = $0.75 / 2\pi$ = 0.12 Hz	A1
1(b)(iii)	$a = r\omega^2$ = $(3.2 \times \frac{2}{3}) \times 0.75^2$	C1
	= 1.2 m s⁻²	A1
1(c)	linear velocity magnitude: X greater than Z	B1
	linear velocity direction: opposite	B1
	angular velocity: magnitude equal <u>and</u> direction same	B1